

CS215: Data Analysis and Interpretation

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Variance

- Let's do a tiny experiment.
 - Please compute the mean and variance
 - $a=[0, 0]$
 - $b=[0, 2]$
 - $c=[0, 4]$

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Sample Variance

Option	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \mu)^2}{n - 1}$	Python (numpy)
0 0	$(0+0)/2 = 0$	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4

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Variance

- Let's continue the tiny experiment.
 - Please compute the mean and variance
 - $a=[0, 0]$
 - $b=[0, 2]$
 - $c=[0, 4]$
 - $d = [2\ 0]$ $e = [2\ 2]$ $f = [2\ 4]$
 - $g = [4\ 0]$ $h = [4\ 2]$ $i = [4\ 4]$

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Sample Variance

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \mu)^2}{n - 1}$	$\frac{\sum (x - \mu)^2}{n}$
0 0	$(0+0)/2 = 0$	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

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Sample Variance

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \mu)^2}{n - 1}$	$\frac{\sum (x - \mu)^2}{n}$
0 0	$(0+0)/2 = 0$	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,

- Dividing by n results in $\bar{s}^2 = \frac{12}{9} = \frac{4}{3}$

- Dividing by (n-1) results in $\bar{s}^2 = \frac{24}{9} = \frac{8}{3}$

- Division by n results in a $\frac{1}{2} \times \frac{8}{3}$

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Example: Pulse-rate of 50 students

Can calculate statistic such as sample mean (79.1) and sample standard deviation (7.6) but we can do more!

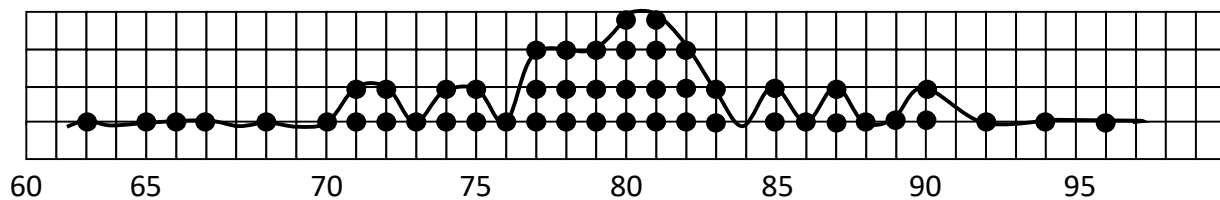
Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let's plot these numbers

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Plot of pulse rate

50 students

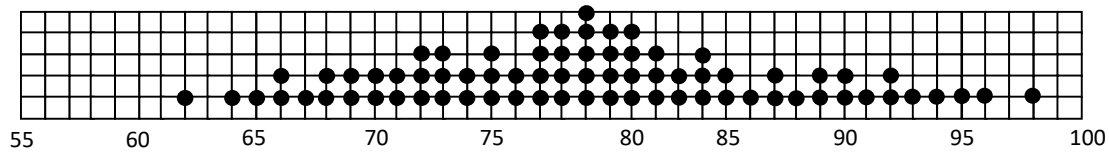


Example from Derek Rowntree

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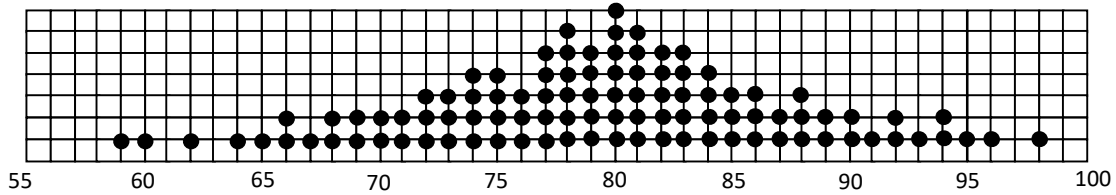
Plot of pulse rate as number increases

75 students



How do we know this?

100 students

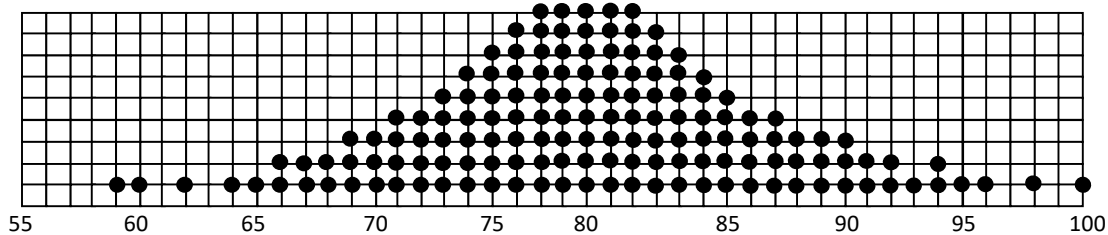


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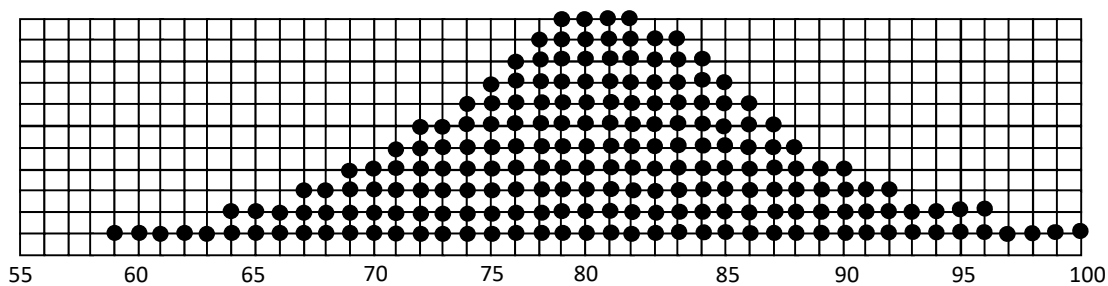
Plot is informally a distribution

The distribution approaches a symmetrical distribution

150 students



200 students



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